

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

V Semester of (BSc) Microbiology Examination April 2018

Course code & Name: MI 502 & MICROBIAL PHYSIOLOGY

Date: 02/04/2018 Day: Monday Time: 01:30 pm To 01:50 pm Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		10
1.	_____ is a sulfur oxidizing bacteria, which deposits sulfur within the cell.		01
	(i) <i>Thiobacillus</i> sp.	(ii) <i>Micromonospora</i> sp.	
	(iii) <i>Beggiatoa</i> sp.	(iv) All of the above	
2.	<i>Gallionella</i> sp. is classified as _____		01
	(i) Sulfur bacteria	(ii) Hydrogen bacteria	
	(iii) Carboxydobacteria	(iv) Iron bacteria	
3.	Heterocyst is produced in response to _____ stress.		01
	(i) Water	(ii) Sunlight	
	(iii) Nitrogen	(iv) Temperature	
4.	_____ inhibits cell division.		01
	(i) ClpX	(ii) Min proteins	
	(iii) ZipA	(iv) Both, i and ii	
5.	_____ possess chlorosomes.		01

	(i) Purple bacteria	(ii) Heliobacteria	
	(iii) Green bacteria	(iv) Blue Green algae	
6.	_____ possess bacteriochlorophyll G.		01
	(i) Purple bacteria	(ii) Heliobacteria	
	(iii) Green bacteria	(iv) Blue Green algae	
7.	Hydrogenases are produced by _____		01
	(i) Hydrogen bacteria	(ii) Iron bacteria	
	(iii) Carboxydobacteria	(iv) Both, i and iii	
8.	Cyanobacteria produce _____ for food storage.		01
	(i) Heterocysts	(ii) Akinetes	
	(iii) Endospore	(iv) None of the above	
9.	_____ is the substrate concentration corresponds to $\frac{1}{2} V_{max}$		01
	(i) $\frac{1}{2} K_m$	(ii) [S]	
	(iii) K_m	(iv) [V]	
10.	_____ utilizes inorganic chemicals as electron donors.		01
	(i) Lithotrophs	(ii) Organotrophs	
	(iii) Autotrophs	(iv) Heterotrophs	

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V Semester of (BSc) Microbiology Examination April 2018

Course code & Name: MI 502 & MICROBIAL PHYSIOLOGY

Date: 02/04/2018 Day: Monday Time: 01:51 pm To 03:30 pm Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I		
Q - II	Answer the following questions as directed. (Any 5)	10
1.	Define: Chemolithotrophy	02
2.	Differentiate: cofactors and coenzymes with a suitable example of each	02
3.	What is photosynthesis?	02
4.	Write a note on Min protein system which regulates cell division.	02
5.	Write a concise note on types of Nitrifying bacteria.	02
6.	Highlight important features of Heliobacteria.	02
7.	What is a biofilm?	02
SECTION - II		
Q-III	Answer the following questions as directed. (Any 3)	15
1.	Explain the process of spore germination	05
2.	Short note: Heterocyst in cyanobacteria	05
3.	Elucidate in details about sulfur oxidizing bacteria	05
4.	What is enzyme inhibition?	05
5.	What is quorum sensing?	05